

Option #2: NLP Chatbot Project Training

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For the Module 5 Portfolio Milestone, the selected project path remains Option #2, the NLP Chatbot Project. This revised milestone follows the dataset strategy selected in the Module 4 Portfolio Milestone: RSiCS for customer-service language patterns, WikiQA for answer-selection and threshold testing, and a sanitized enterprise FAQ/SOP corpus for the domain-specific knowledge base. The Module 5 training run focused on the custom FAQ/SOP corpus because it is the part of the data strategy that contains approved route labels, knowledge-base wording, and response text.

Model Type and Dataset

The first trained model was a retrieval-based TF-IDF cosine-similarity baseline. This model type is a better match for the Module 4 plan than the previous classifier because the proposed chatbot is intended to retrieve or route to approved knowledge-base content instead of generating open-ended answers. TF-IDF converts the FAQ/SOP entries into weighted term vectors, and cosine similarity ranks the closest approved entry for a new support message. This follows the general information-retrieval approach described by Manning et al. (2008).

The actual training file was a sanitized enterprise FAQ/SOP corpus created for this coursework prototype. It contains 80 entries across 10 route labels, including account access, billing and invoice, order status, returns and refunds, service scheduling, technical support, policy lookup, onboarding, training materials, and human escalation. Each row includes a sample question, keywords, approved answer, category, route label, and a source note. RSiCS and WikiQA were not used as direct training inputs in this first run; instead, they remain the planned supplemental datasets. RSiCS can later test conversational customer-service language variation (Beaver et al., 2017), while WikiQA can later test answer ranking and no-answer threshold behavior (Yang et al., 2015).

Hyperparameters and Training Setup

The final training run used 50 training rows and 30 test rows across 10 route labels. The split was grouped by source row so that each FAQ/SOP entry remained in either training or testing. The held-out group ratio was 0.40, and the random seed was 525. This keeps the evaluation reproducible while still holding back examples from every route.

The final representation used the combined_text field, lowercase 1-1 n-gram TF-IDF vectors, and a vocabulary of 212 features. The maximum feature limit was 2500, and the draft similarity threshold was 0.20. The selected text field combines the user's sample question with curated keywords because the Module 4 plan framed the custom dataset as a managed FAQ/SOP knowledge base, not as unlabeled raw chat logs.

Training Results

The first formal training run produced an accuracy of 1.0000, macro precision of 1.0000, macro recall of 1.0000, macro F1 score of 1.0000, and mean similarity of 0.7323. No test entry fell below the 0.20 similarity threshold. This strong result shows that the retrieval model can separate the controlled FAQ/SOP routes when each entry includes approved keywords and domain-specific support language.

The result should be interpreted carefully. A hyperparameter sweep showed that the best question-only run reached only 0.2333 accuracy and 0.2094 macro F1. That contrast is important: the controlled FAQ/SOP metadata made the first retrieval model work, but raw customer language still needs more examples and broader variation. The high score is therefore a proof that the knowledge-base structure is usable, not proof that the chatbot is production-ready.

Reflection and Next Steps

This redo is more consistent with the prior portfolio plan because it trains against the intended enterprise knowledge-base format. The training experience also clarified a key project risk: a retrieval chatbot depends heavily on the quality of its curated entries. When route labels, approved answers, and keywords are clean, the model can route reliably. When only short customer questions are available, the same small corpus does not provide enough lexical coverage for dependable routing.

The next step is to expand the custom corpus and then test it against the two public datasets selected in Module 4. RSiCS should be used to add realistic customer-service phrasing, including relational or emotional language that may surround the core request. WikiQA should be used to test whether a candidate answer is relevant enough to return or whether the chatbot should ask a clarifying question. This preserves the business-safe design: high-confidence matches return approved content, uncertain matches ask for clarification, and low-confidence cases escalate to a human agent.

Use of AI Assistance

AI assistance was used to help structure the training script, organize the milestone files, and draft this report. The model was executed locally, and the reported metrics, validation checks, and output files were generated from the local CSC525 workspace before submission.

References

- Beaver, I., Freeman, C., & Mueen, A. (2017). An annotated corpus of relational strategies in customer service. arXiv. <https://doi.org/10.48550/arXiv.1708.05449>
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- Yang, Y., Yih, W.-t., & Meek, C. (2015). WikiQA: A challenge dataset for open-domain question answering. In L. Marquez, C. Callison-Burch, & J. Su (Eds.), Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing (pp. 2013-2018). Association for Computational Linguistics. <https://doi.org/10.18653/v1/D15-1237>